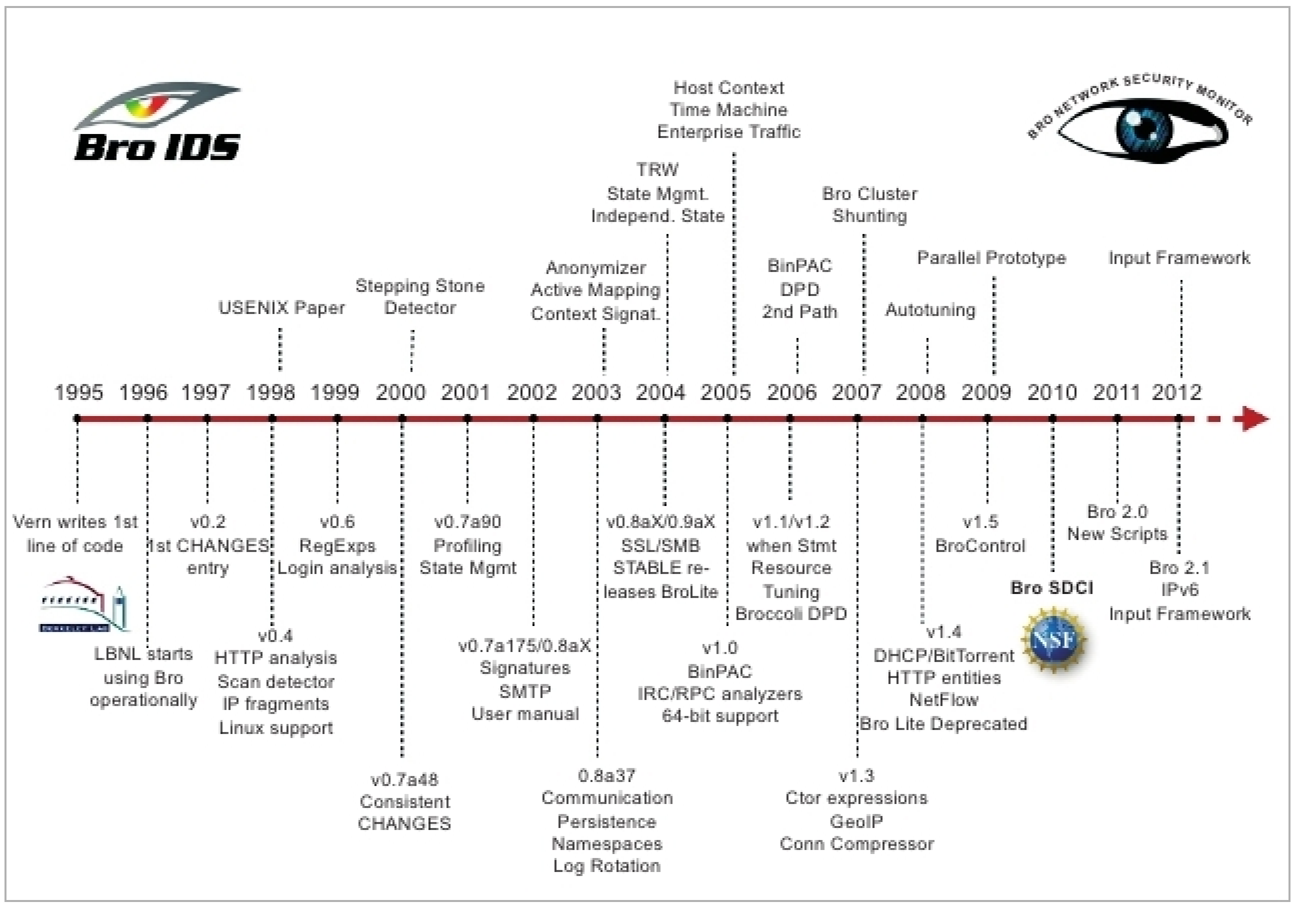

Figure 2: History of Zeek

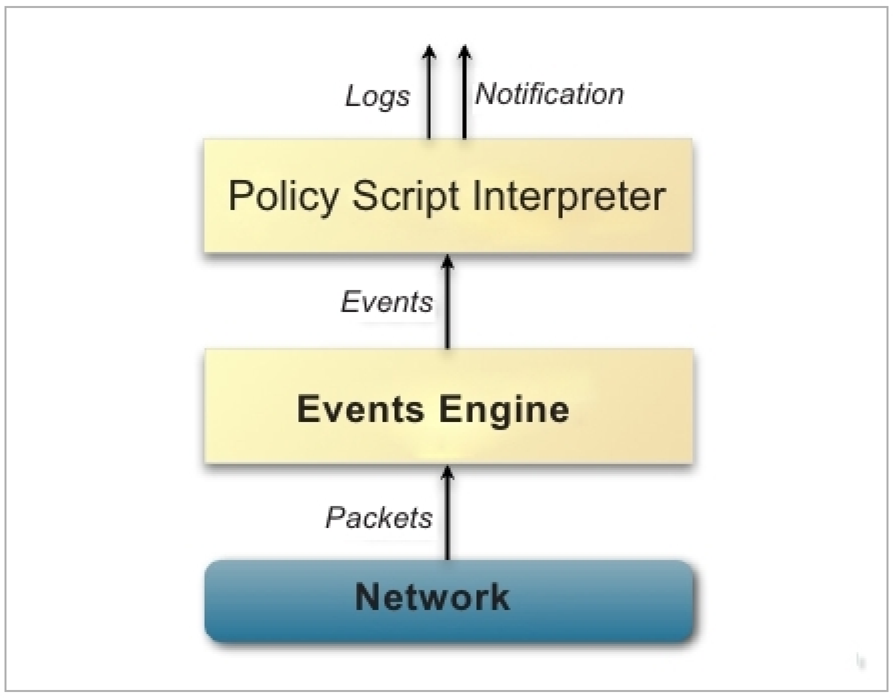

Figure 3: The Zeek architecture

- **In-depth analysis:** The analysis is provided by a variety of analysers and is hence in-depth.
- **Open interfaces:** Zeek provides open interfaces for exchanging real-time information.
- **Open Source:** It is released under the BSD licence. This makes it easier and effective to use Zeek for various scenarios.

History and architecture

The evolution of Zeek is shown in Figure 2 (you can refer to Zeek's official documentation).

The earliest work on Zeek dates back to 1995. Over 20 years of research has made Zeek very powerful and effective. Excellent contributions by people from both academia and industry have made Zeek a robust security mechanism.

Zeek is designed with a two-layer architecture, as shown in Figure 3 (the source is official documentation).

Event engine: The first layer of Zeek is the event engine that observes the incoming packet stream and converts it into a series of events. Based on the nature of the incoming packet sequence, an event is detected. For example, if an HTTP request is raised, it converts it into an *http_request* event. The raised *http_request* incorporates the following data items along with it:

- The IP address
- The port address
- The request URI
- The HTTP version information.

The role of the event engine is to simply convert the incoming sequence into an event. The event engine doesn't associate any interpretation to the detected event.

Script interpreter: The script interpreter receives the input from the event handler. The role of the script interpreter is to interpret the events detected by the event engine. The script interpreter executes the respective event handlers that are built in Zeek's scripting language. Hence, the script interpreter is the most important component that enables the generation of real-time alerts.

Installation

To install Zeek in your system, the following dependencies need to be satisfied:

- Lipcap
- Open SSL libraries
- Libz
- BIND8 library
- Python 2.6 or above

Run the following command to install Zeek:

```
sudo apt-get install cmake make gcc g++ flex bison libpcap-dev
libssl-dev python-dev swig zlib1g-dev
```

Primarily, Zeek requires a UNIX platform. Support is provided for Linux, FreeBSD and MacOS X. The Linux binaries can be downloaded from the official link: *https://www.zeek.org/download/index.html.* (The stable release is Zeek 2.6.1, which was released in December 2018.)

In addition to the binaries, Zeek can be compiled from source as well. Detailed instructions and the dependencies list can be collected from *https://docs.zeek.org/en/stable/install/install.html.*